Health and Safety at Work

'Personal Protective Equipment (PPE)'

Learning aims

You should be able to:

Identify the factors that influence the choice of ***Personal Protective Equipment (PPE)*** for given situations, to include:

- chemical
- metal working fluids
- temperature
- crush resistance
- noise
- fume
- eye
- electrical
- radiation protection

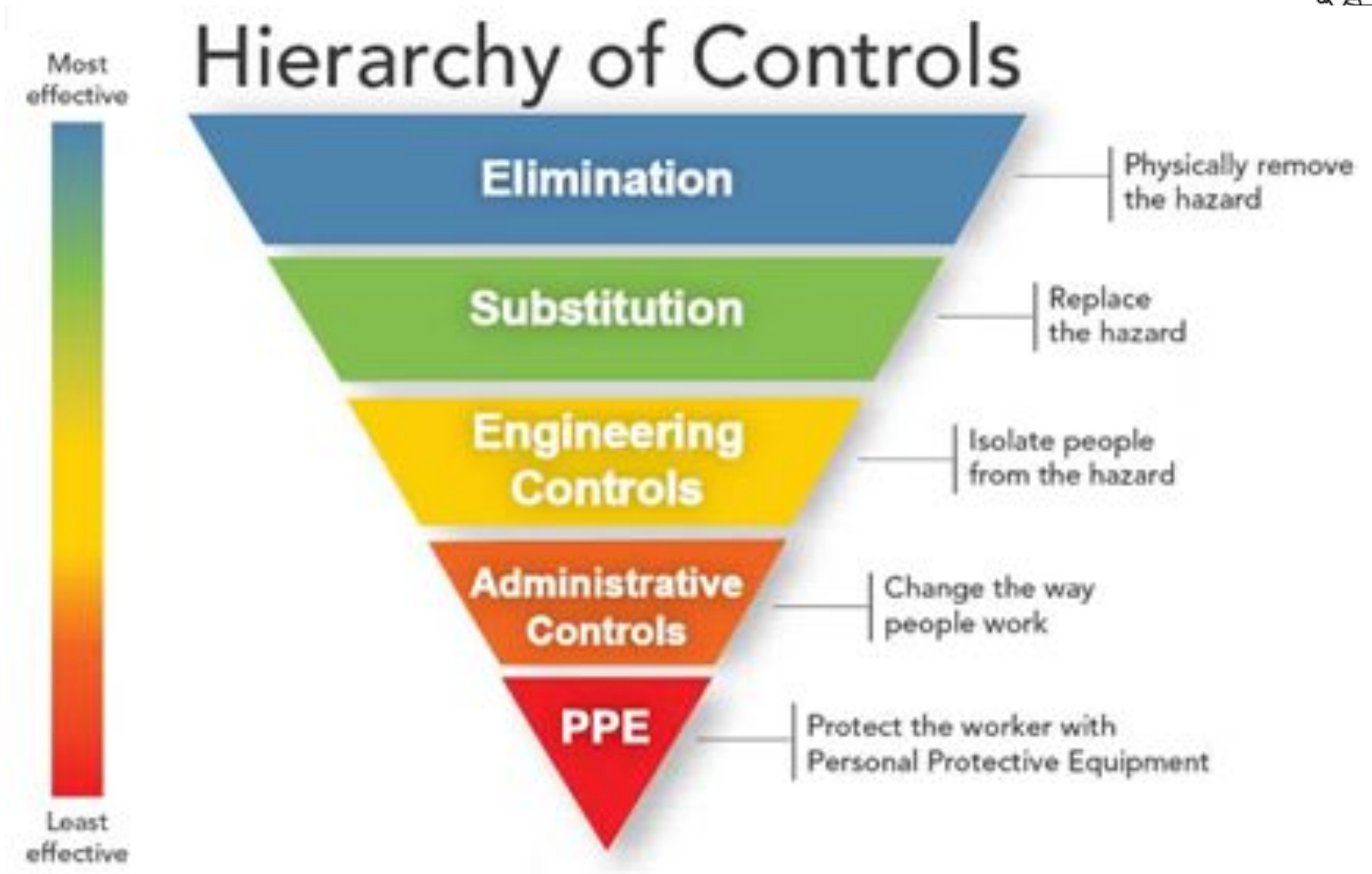
Hierarchy of Control

Consider this before we start …

Hierarchy of Control

Occupational safety and health professionals use the hierarchy of control (shown in the next slide) to determine how to implement feasible and effective controls. This approach groups actions by their likely effectiveness in reducing or removing the hazard.

Hierarchy of Control



Hierarchy of Control Cont.

Note: the use of PPE is the least effective control measure and is implemented after other methods have been considered.

Health and Safety at Work

'Personal Protective Equipment (PPE)'

Personal Protective Equipment

- *Introduction*
- *Eye & Face Protection*
- *Head Protection*
- *Hand Protection*
- *Foot Protection*
- *Body Protection*

Introduction

PPE regulations 2002

This standard requires that employers must provide at no cost to their employees PPE when such equipment is necessary to protect employees from job related injuries, illnesses and fatalities

Protecting Employees from Workplace Hazards

Employers *must* protect ***employees*** from workplace hazards and dangerous work procedures that can cause injury, illness and fatalities.

Employers must:

- *Use all feasible engineering and work practice controls to eliminate and reduce hazards*
- *Then use appropriate PPE if these controls do not eliminate hazards*

Remember: PPE is the last control

LOOK AND SEE…

Look at the next ***four*** slides – do you think these are appropriate forms of PPE???

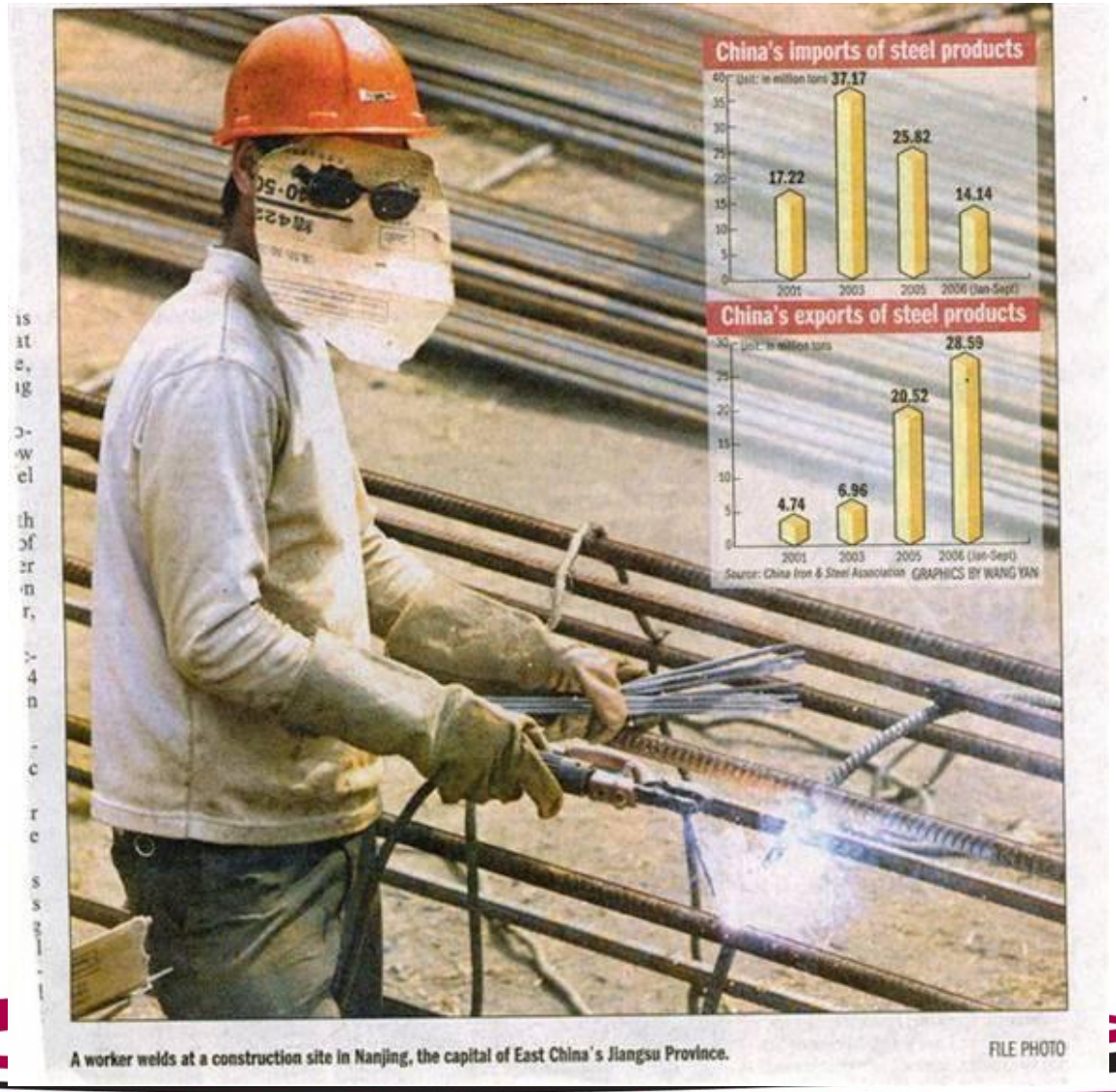
Is this an appropriate ***Hard Hat***?



Is this an appropriate ***Face Shield***?



Is this an appropriate ***Welder's Mask***?



A worker welds at a construction site in Nanjing, the capital of East China's Jiangsu Province. FILE PHOTO

Is this an appropriate ***Scaffolding***?



Training

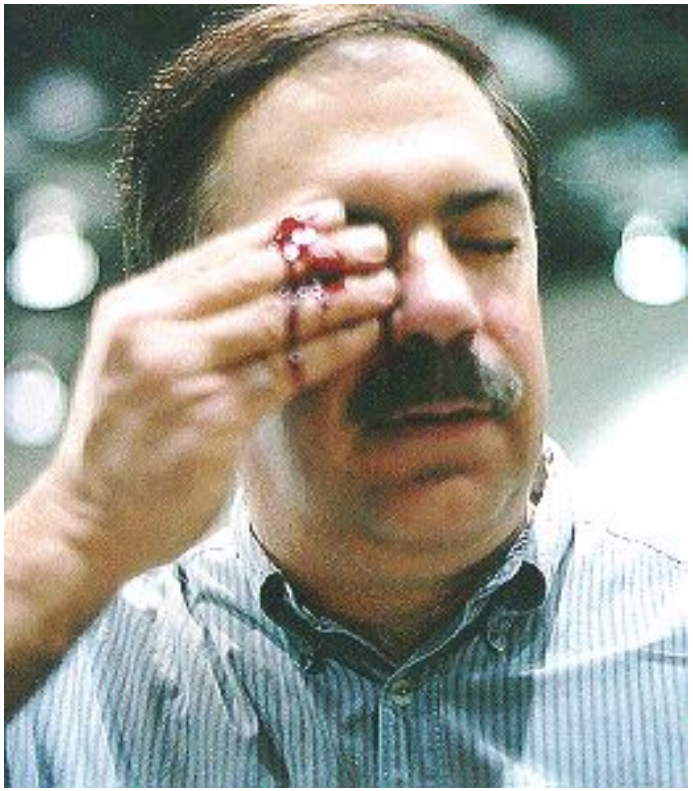
Employees required to use PPE must be trained to know at least the following:

- *When PPE is necessary*
- *What type of PPE is necessary*
- *How to properly put on, take off, adjust, and wear*
- *Limitations of the PPE*
- *Proper care, maintenance, useful life and disposal*

Eye and Face Protection

Why Eye and Face Protection is Important???

Thousands of people are blinded each year from work related eye injuries. Injuries that could have been prevented, if only people would have used eye or face protection.



Eye and Face Protection

Types of eye/face hazards

- *Impact*
- *Heat*
- *Chemicals*
- *Dust*
- *Light and/or Radiation*



Eye and Face Protection

Types of Eye and Face Personal Protective Equipment

Safety Glasses:



- *much stronger and more resistant to impact and heat than regular glasses*
- *equipped with side shields that give you protection from hazards that may not be directly in front of you*
- *Limitation - Does not seal around eyes, could allow small droplets to come in contact with eyes*

Eye and Face Protection





Regular glasses or sunglasses are not appropriate ***SAFETY GLASSES***

Eye and Face Protection

Types of Eye and Face Personal Protective Equipment

Goggles:



- *surround the eye area, they give you more protection in situations where you might encounter splashing liquids, fumes, vapours, powders, dusts, and mists*
- *must indicate that they are chemical splash goggles to be worn for that purpose*
- *Limitation - Uncomfortable to wear with other head gear like helmet, ear muffs or respirator*

Eye and Face Protection

Types of Eye and Face Personal Protective Equipment

Face Shields:



- *full face protection*
- *used around operations which expose you to molten metal, chemical splashes, or flying particles*

Limitations:

- *Are not considered eye protection, will need to wear goggles or glasses underneath*
- *Can fog up if working in poorly ventilated area*

Eye and Face Protection

Types of Eye and Face Personal Protective Equipment

Welding Helmets:

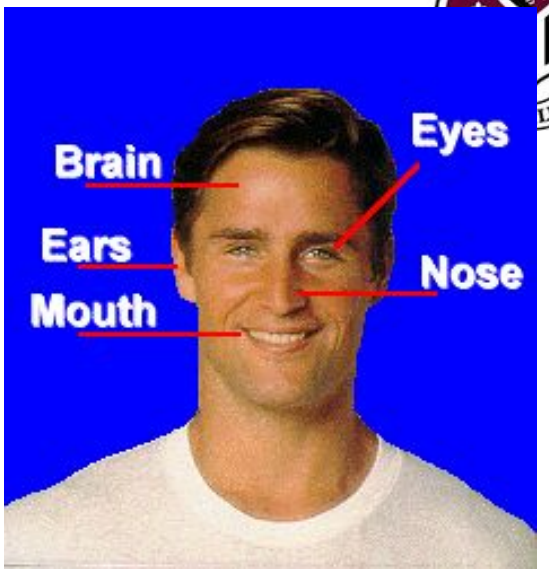


- *provide both face and eye protection*
- *use special absorptive lenses that filter the intense light and radiant energy that is produced during welding operations*

Limitations:

- *Heavy and hot, fog up*
- Must wear safety glasses or goggles underneath helmet

Head Protection



Why is Head Protection Important?

In and around your head are:

- *Your eyes, with which you see*
- *Your ears, with which you hear*
- *Your nose, with which you smell*
- *Your mouth, with which you eat and speak*
- *Your brain, with which you think*

Injuries to the head are very serious. For this reason, head protection and safety are very important .

Head Protection

Types of ***Head hazards***:

- ***Impact***
- ***Electric shock***
- ***Drips***



Head Protection

Types of Head PPE



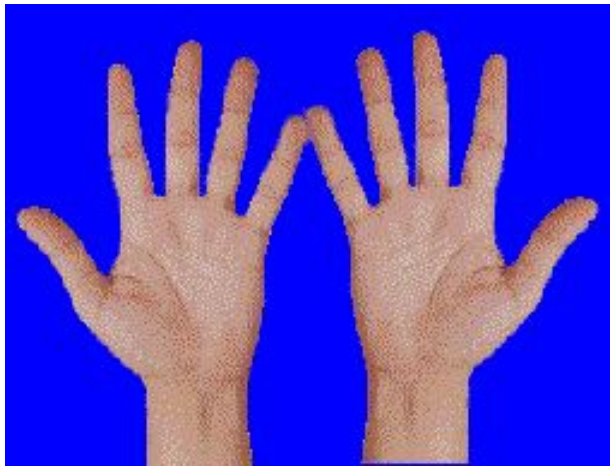
Hard hats

- *A rigid shell that resists and deflects blows to the head*
- *A suspension system inside the hat that acts as a shock absorber*
- *Some hats serve as an insulator against electrical shocks*
- *Shields your scalp, face, neck, and shoulders against splashes, spills, and drips*
- *Some hard hats can be modified so you can add face shields, goggles, hoods, or hearing protection to them*

Hand Protection

Why is Hand Protection Important?

- ***It has been estimated that almost 20% of all disabling accidents on the job involve the hands***
- *Without your fingers or hands, your ability to work would be greatly reduced*



Hand Protection

Proper Fit and Use of PPE

- *Select and use the right kind of glove for the job you are going to be performing*
- *Check fit, always use correct size*
- *Make sure chemical resistant to chemical being used*
- *Inspect your gloves before you use them*
- *Remove any rings, watches, or bracelets that might cut or tear your gloves*
- *Look for holes and cracks that might leak*
- *Replace gloves that are worn or torn*



Foot Protection

Why is Foot Protection Important?

- *The human foot is rigid enough to support the weight of your entire body, and yet flexible enough to allow you to run, dance, play sports, and to take you anywhere you want to go. Without your feet and toes, your ability to work at your job would be greatly reduced.*



Foot Protection

Types of Foot Hazards:

- ***Impact Injuries***
- ***Injuries from Spills and Splashes***
- ***Compression Injuries***
- ***Electrical Shocks***
- ***Extremes in Cold, Heat, and Moisture***
- ***Slipping***

Foot Protection





Potential Incidences of Foot Hazards

Impact Injuries:

- *At work, heavy objects can fall on your feet. If you work around sharp objects, you can step on something sharp and puncture your foot*

Injuries from Spills and Splashes:

- *Liquids such as acids, caustics, and molten metals can spill into your shoes and boots. These hazardous materials can cause chemical and heat burns*

Foot Protection









Types of Foot PPE

Safety Shoes and Boots

Steel toe footwear

- *protects your toes from falling objects and from being crushed*

Body Protection

Why is Body Protection Important?

- *The skin acts as a natural barrier to the elements*
- *Chemicals can break down the skin barrier and allow secondary infections to manifest*



Body Protection

Types of Body Hazards:

- ***Temperature stress***
- ***Chemical Contact***
- ***Radiation***

Body Protection

Types of Body Protection

Coveralls

- *Chemical rated*



Full body suit

- *Hazardous materials handling*
- *Carbon filtering for emergency response*

